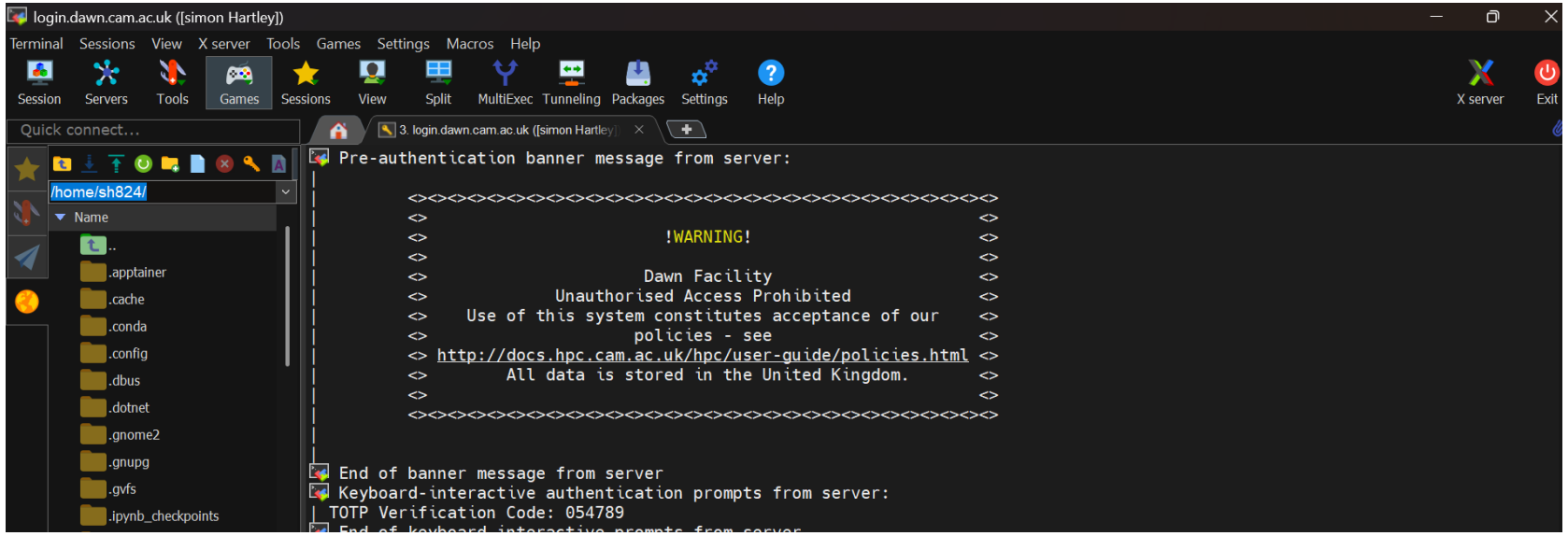
Intro to Dawn

RSE DAWN TEAM

2. Connecting to DAWN

Connect using an SSH Client.

Windows: PuTTY | MobaXterm | Windows Subsystem for Linux
MacOS or Linux: Terminal



Projects and QoS

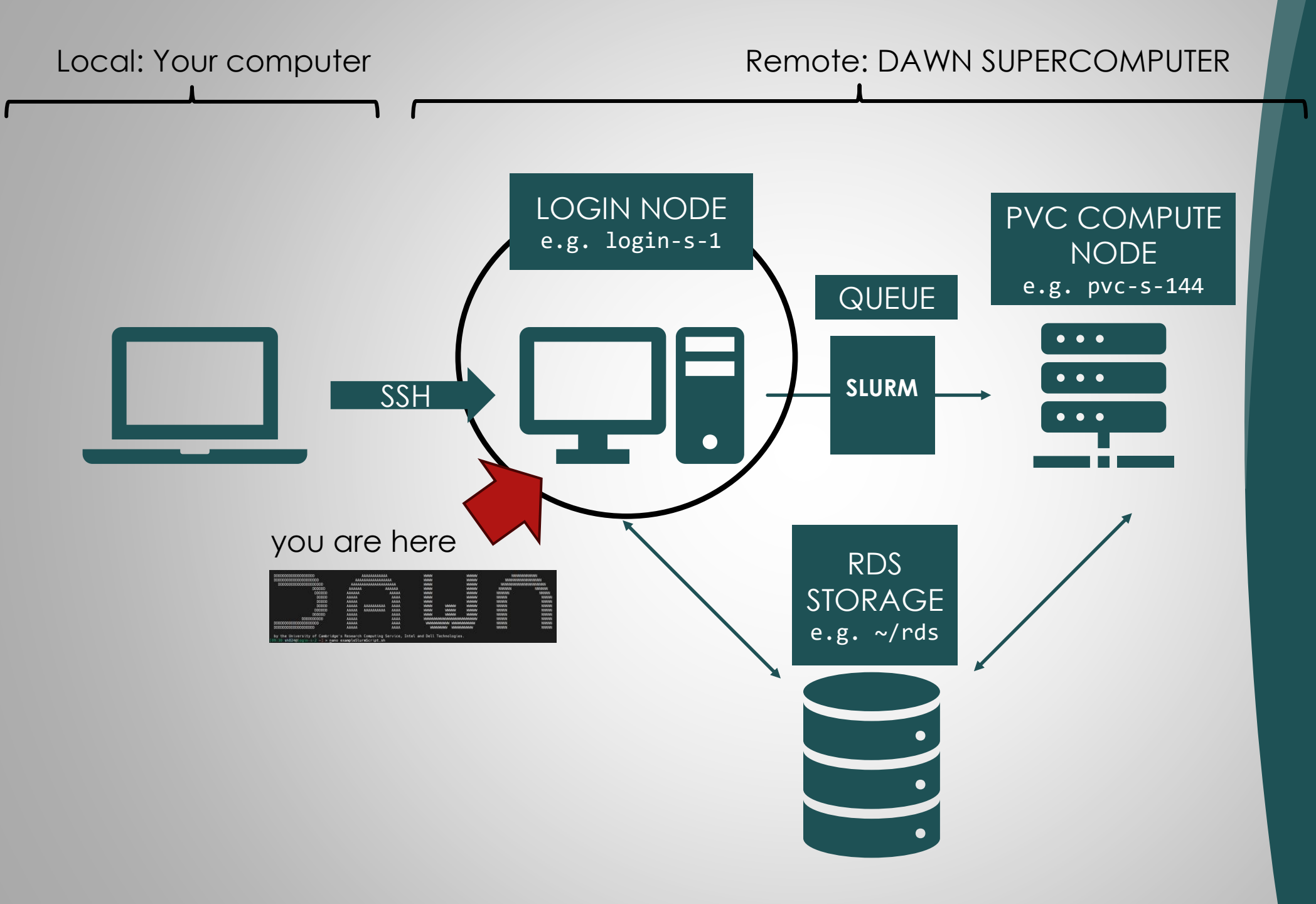
Once you are connected to a login node, run `mybalance`

```
[10:06 sh824@login-s-2 ~] > mybalance
User           Usage |        Account     Usage | Account Limit Available (hours)
---------- -------- + -------------- -------- + ------------- ---------
sh824             0 |    RCS-RSE-CPU         0 |       100,000   100,000
sh824             0 | RCS-RSE-DAWN-GPU       1 |        10,000     9,999
```

to see information about your DAWN projects and available QoS.

This information is more detailed at
https://access.hpc.cam.ac.uk/projects/

System Overview



Your Home folder:50GB

Once you are connected to a login node, run `quota`

```
[14:19 sh824@login-s-2 ~] > pwd -P
/home/sh824
[14:24 sh824@login-s-2 ~] > quota
Filesystem/Project    GB        quota     limit          grace           files    quota    limit   grace User/Grp/Proj
/home                  9.5       50.0      55.0                    -    ------- No File Quotas  ------- U:sh824
/rds-d6              755.5        0.0       0.0                    -  1013171        0        0       - G:sh824
/rds-d7                0.0        0.0       0.0                    -        4        0        0       - G:sh824
rds-Om6DXPhIBgg       67.0     5000.0    5000.0                    -   424979  2560000  2560000       - P:93626
rds-rgtMZbbbjkQ    15029.4    16000.0   16000.0                    -   332035  8192000  8192000       - P:92249
```

Your Project folder: 2TB>

Once you are connected to a login node, run `quota`

```
[14:19 sh824@login-s-2 ~] > pwd -P
/home/sh824
[14:24 sh824@login-s-2 ~] > quota
Filesystem/Project    GB        quota     limit          grace           files    quota    limit   grace User/Grp/Proj
/home                  9.5       50.0      55.0                    -    ------- No File Quotas  ------- U:sh824
/rds-d6              755.5        0.0       0.0                    -  1013171        0        0       - G:sh824
/rds-d7                0.0        0.0       0.0                    -        4        0        0       - G:sh824
rds-Om6DXPhIBgg       67.0     5000.0    5000.0                    -   424979  2560000  2560000       - P:93626
rds-rgtMZbbbjkQ    15029.4    16000.0   16000.0                    -   332035  8192000  8192000       - P:92249
```

Choose your editor

Once you are connected to a login node, run your editor

- nano

```
                    :::
  iLE88Dj.  :jD88888Dj:
.LGitE888D.f8GjjjL8888E;
iE   :8888Et.     .G8888.
;i    E888,        ,8888,
      D888,        :8888:
      D888,        :8888:
      D888,        :8888:
      D888,        :8888:
      888W,        :8888:
      W88W,        :8888:
      W88W:        :8888:
      DGGD:        :8888:
                   :8888:
                   :W888:
                   :8888:
                    E888i
                    tW88D
```

- vim

```
[09:41 sh824@login-s-2 ~] > vim exampleSlurmScript.sh
```

```
[09:30 sh824@login-s-2 ~] > nano exampleSlurmScript.sh
```

Create your job script

```
#!/bin/bash -l
# --- account -----------------------------------------------------
#SBATCH --account=FIX_ME      # Only required if you are a member of more than one Dawn project
#SBATCH --partition=pvc9
#SBATCH --qos=FIX_ME          # upon signing-up to Dawn you will be assigned a qos

# --- resources ---------------------------------------------------
# Slurm to send a signal to our script 120
# job name
#SBATCH --job-name=example
#SBATCH --time=0-01:00:00
#SBATCH --nodes=1                   # node count Normally set to 1
#SBATCH --ntasks-per-node=1         # total number of tasks per node
#SBATCH --gres=gpu:1                # number of allocated gpus per node

# --- environment -------------------------------------------------
module purge
module load rhel9/default-dawn

# Visit https://github.com/RSE-Cambridge/Intro-to-Dawn for more information

# Example 1 - This example just shows the bare minimum of module loading with a job asking for 1 hour of time.
# It prints "Script is running on <HOSTNAME>" then waits for 15s so you can see it move through status changes in the squeue command.
# You should see that the hostname printed into the slurm*.out file is different to that in your command line prompt

#Use 'bash remove_slurm_files.sh' to delete the slurm files before moving onto the next example

module purge
module load rhel9/default-dawn

echo -n "Script is running on "
hostname
sleep 15
```

Submit your job script

```
(10:03 sh824@pvc-s-144 ~) > sbatch ./exampleSlurmScript.sh
Submitted batch job 24556024
```

Check your job status with squeue

```
(10:03 sh824@pvc-s-144 ~) > squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          24556024      pvc9  example    sh824 PD       0:00      1 (None)
```

```
       JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
    24556170      pvc9  example    sh824  R       0:01      1 pvc-s-144
```

Check your job outputs

```
[10:15 sh824@login-s-2 ~] > ls -lathr slurm-*.out
-rw-r--r-- 1 sh824 sh824 233 Mar  9 10:04 slurm-24556024.out
-rw-r--r-- 1 sh824 sh824 233 Mar  9 10:07 slurm-24556170.out
-rw-r--r-- 1 sh824 sh824 233 Mar  9 10:08 slurm-24556276.out
-rw-r--r-- 1 sh824 sh824 233 Mar  9 10:10 slurm-24556389.out
-rw-r--r-- 1 sh824 sh824 233 Mar  9 10:10 slurm-24556396.out
[10:15 sh824@login-s-2 ~] > cat ./slurm-24556024.out
Loading rhel9/default-dawn
  Loading requirement: rhel9/global rhel9/slurm dawn-env-rhel9/2025-03-23
Loading rhel9/default-dawn
  Loading requirement: rhel9/global rhel9/slurm dawn-env-rhel9/2025-03-23
Script is running on pvc-s-144
[10:17 sh824@login-s-2 ~] >
```

Experiment with your job

- Add the line `sleep 5h` before the `echo` in your job script and submit your job. How do you cancel the job? *Hint:* use the command `scancel <jobid>`
- What happens if you request a wall-time job of 1 minute and `sleep 120` ?

https://github.com/RSE-Cambridge/Intro-to-Dawn

Experiment with your job

```
(11:38 sh824@pvc-s-144 Intro-to-Dawn) >  squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          24568982      pvc9 example3    sh824  R       0:19      1 pvc-s-144
          24553214      pvc9 sys/dash    sh824  R    2:04:50      1 pvc-s-144
(11:38 sh824@pvc-s-144 Intro-to-Dawn) > scancel 24568982
(11:39 sh824@pvc-s-144 Intro-to-Dawn) >  squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          24553214      pvc9 sys/dash    sh824  R    2:05:41      1 pvc-s-144
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Experiment with your job

- What happens if you **sleep 120** and request a wall-time job of 1 minute?

```
(11:48 sh824@pvc-s-144 Intro-to-Dawn) > sbatch ./3-timeout.sh
Submitted batch job 24569975
(11:49 sh824@pvc-s-144 Intro-to-Dawn) >  squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          24569975      pvc9 example3    sh824  R       0:02      1 pvc-s-144
          24553214      pvc9 sys/dash    sh824  R    2:15:52      1 pvc-s-144
(11:49 sh824@pvc-s-144 Intro-to-Dawn) > cat slurm-24569857.out
Loading rhel9/default-dawn
  Loading requirement: rhel9/global rhel9/slurm dawn-env-rhel9/2025-03-23
slurmstepd: error: *** JOB 24569857 ON pvc-s-144 CANCELLED AT 2026-03-09T11:48:57 DUE TO TIME LIMIT ***
(11:49 sh824@pvc-s-144 Intro-to-Dawn) >
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Interactive Jobs

- The command `sinter` is a shortcut for interactive jobs:

```
[11:53 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] >

[11:54 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] >
[11:54 sh824@login-s-2 rds-aswsai] > sintr --account=support-gpu \
      --partition=pvc9 \
      --nodes=1 \
      --ntasks-per-node=1 \
      --cpus-per-task=36 \
      --mem-per-cpu=3800 \
      --gres=gpu:4 \
      --time=01:00:00 \
         "$@" \
      --qos=INTR \
      --mail-type=start,end \
      --mail-user=sh824@cam.ac.uk \
      --job-name=job-$(date +"%Y_%m_%d_%I_%M")
[11:54 sh824@login-s-2 rds-aswsai] >
```

```
[screen 1: bash] @login-s-2

(11:54 sh824@pvc-s-247 rds-aswsai) >
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Interactive Jobs

```
[11:54 sh824@login-s-2 rds-aswsai] > squeue -u $USER
             JOBID PARTITION     NAME     USER ST       TIME  NODES NODELIST(REASON)
          24570284      pvc9 _interac    sh824  R       2:00      1 pvc-s-247
          24553214      pvc9 sys/dash    sh824  R    2:22:31      1 pvc-s-144
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Interactive Jobs

- The command `sinter` is a shortcut for interactive jobs:

```
sintr --account=FIX_ME --partition=pvc9  --nodes=1 --ntasks-per-node=1
--cpus-per-task=36 --mem-per-cpu=3800 --gres=gpu:4 \
      --time=01:00:00 \
          "$@" \
      --qos=INTR \
      --mail-type=start,end \
      --mail-user=<your_email> \
      --job-name=job-$(date +"%Y_%m_%d_%I_%M")
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Accessing Software Applictions

- The Look at your Launch Script `1-basicSlurm.sh`:

```
# --- environment ----------------------------------------------------
module purge
module load rhel9/default-dawn
```

- Here we load default software

https://github.com/RSE-Cambridge/Intro-to-Dawn

Accessing Software Applictions

- The Look for software on the system:

```
[12:02 sh824@login-s-2 Intro-to-Dawn] > nano 1-basicSlurm.sh
[12:27 sh824@login-s-2 Intro-to-Dawn] > module avail
  1 ------------------
```

Accessing Software Applictions

- Filter for software on the system:

```
[12:31 sh824@login-s-2 Intro-to-Dawn] > module purge; module load rhel9/default-dawn
Loading rhel9/default-dawn
  Loading requirement: rhel9/global rhel9/slurm dawn-env-rhel9/2025-03-23
[12:31 sh824@login-s-2 Intro-to-Dawn] > module avail
------------------
```

Accessing Software Applictions

- Filter for Python software on the system:

```
[12:35 sh824@login-s-2 Intro-to-Dawn] > module avail *python*
      1 ----------------------------------------------------------------------
      1 --------
      2 intelpython-conda/2024.0.1.3(default)  intelpython-conda/2025.0
      3
      4 Key:
      5 (symbolic-version)
[12:35 sh824@login-s-2 Intro-to-Dawn] > module load intelpython-conda/2025.0
(intelpython) [12:37 sh824@login-s-2 Intro-to-Dawn] > which python
/usr/local/dawn/software/external/intel-oneapi-ai/2025.0/intelpython/bin/python
(intelpython) [12:37 sh824@login-s-2 Intro-to-Dawn] >
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

View loaded Modules

- Modules Loaded on the system:

```
(intelpython) [12:37 sh824@login-s-2 Intro-to-Dawn] > module list
Currently Loaded Modulefiles:
 1) rhel9/global <aL>   2) rhel9/slurm <aL>   3) dawn-env-rhel9/2025-03-23 <aL>   4) rhel9/default-dawn   5) intelpython-conda/2025.0

Key:
<module-tag>  <aL>=auto-loaded
(intelpython) [12:38 sh824@login-s-2 Intro-to-Dawn] >
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Your turn

- Create a Python script called **hello.py** that contains:

```python
# hello.py

import time
import socket

node=socket.gethostname();
ts1=time.ctime()
printf(f"hello from {node} at  {ts1}")
time.sleep(5)
ts2=time.ctime()
printf(f"ByeBye from {node} at {ts2}")
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Your turn

- **hello.py** Outputs:

```
(intelpython) [13:07 sh824@login-s-2 Intro-to-Dawn] > python hello.py
hello from login-s-2 at  Mon Mar  9 13:07:38 2026
ByeBye from login-s-2 at Mon Mar  9 13:07:43 2026
(intelpython) [13:07 sh824@login-s-2 Intro-to-Dawn] >
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Slurm Array Jobs

- Run many jobs from one Slurm file
- `#SBATCH --array=1-10%2`

Adding this will run the script 10 times with 2 jobs running simultaneously

Need to use with environment variables to make it useful.

- `SLURM_JOB_ID` - Full job number for each array job
- `SLURM_ARRAY_JOB_ID` - Job id of the array itself
- `SLURM_ARRAY_TASK_ID` - Index of the job in the array, e.g. a value between 1 and 10 from above.

To track these jobs, use the `sacct -j <Job id>` command or the `squeue` command.

https://github.com/RSE-Cambridge/Intro-to-Dawn

Slurm Array Jobs

+ How do I give different values to each job?
  - Use the SLURM_ARRAY_TASK_ID as a lookup to the information you want to use.

+ How do I make a Slurm array bigger than MaxArraySize?
  - Set to 4000 on DAWN, defaults to 1001.
  - Use a script to repeatedly launch a slurm job with an increasing offset each time 1,2,3.....
    * Slurm scripts can take command line arguments
    * Add this to SLURM_ARRAY_TASK_ID before doing the lookup.

https://github.com/RSE-Cambridge/Intro-to-Dawn

Slurm signal

- + How do I save my job before it jets killed?

```bash
#!/bin/bash
# --- account ------------------------------------------------------
#SBATCH --qos FIX_THIS
#SBATCH --account FIX_THIS
#SBATCH --partition=pvc9

# --- resources ----------------------------------------------------
# Slurm to send a signal to our script 120
# job name
#SBATCH --job-name=example
#SBATCH --time=0-00:04:00
#SBATCH --signal=B:USR1@120
#SBATCH --nodes=1                    # node count Normally set to 1
#SBATCH --ntasks-per-node=2          # total number of tasks per node
#SBATCH --gres=gpu:1                 # number of allocated gpus per node

your_cleanup_function()
{
    echo "function your_cleanup_function called at $(date)"
    # do whatever cleanup you want here
}

# call your_cleanup_function once we receive USR1 signal
trap 'your_cleanup_function' USR1

echo "starting calculation at $(date)"

# the calculation "computes" (in this case sleeps) for 1000 seconds
# but we asked slurm only for 240 seconds so it will not finish
# the "&" after the compute step and "wait" are important
sleep 1000 &
wait
```

https://github.com/RSE-Cambridge/Intro-to-Dawn

Slurm Job Dependancies Jobs

- + How do I set one job to depend on another?

```
#!/bin/bash
# --- first job - no dependencies ----------------------------
# --- Build the SIF File -------------------------------------
jid1=$(sbatch --parsable 2.1_build_INTEL-xpu-smi.sbatch)

# --- second task of our pipeline ----------------------------
# --- use the previous variable from the SLURMJOBID here -----
# --- Run the SIF File ---------------------------------------
sbatch --dependency afterok:${jid1} 2.2_run_INTEL-xpu-smi.sbatch
# second submission in case the first one fails
run2_id=$(sbatch --parsable --dependency afternotok:${jid1} 3.0_clean.sh)
```

1.0_buildrun	09/03/2026 12:49	SH File	1 KB
2.1_build_INTEL-xpu-smi	09/03/2026 12:53	SBATCH File	3 KB
2.2_run_INTEL-xpu-smi	09/03/2026 12:53	SBATCH File	3 KB
3.0_clean	09/03/2026 12:49	SH File	1 KB
INTEL-xpu-smi	09/03/2026 12:49	DEF File	2 KB

https://github.com/RSE-Cambridge/Intro-to-Dawn

When looking for answers try these pages:

dawn-intel-gpu-pvc-nodes

Thank You

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